Dept_ID	Dept_Name	Location
1	HR	Hyderabad
2	IT	Bangalore
3	Finance	Chennai
4	Marketing	Mumbai
5	Sales	Delhi

Em...	Emp_...	Age	Salary	Dept_ID
101	Rahul	25	300...	1
102	Priya	27	450...	2
103	Arjun	29	500...	2
104	Sneha	24	280...	3
105	Kiran	30	550...	4
106	Ananya	26	350...	1
107	Ravi	32	600...	5
108	Meena	28	420...	3
109	Vikram	31	480...	4
110	Divya	23	250...	5

```sql
SELECT Emp_id, Emp_name FROM Employee;
```

Emp_ID	Emp_Name
101	Rahul
102	Priya
103	Arjun
104	Sneha
105	Kiran
106	Ananya
107	Ravi
108	Meena
109	Vikram
110	Divya

```sql
SELECT * FROM employee WHERE salary >=40000;
```

Em...	Emp_...	Age	Salary	Dept_ID
102	Priya	27	450...	2
103	Arjun	29	500...	2
105	Kiran	30	550...	4
107	Ravi	32	600...	5
108	Meena	28	420...	3
109	Vikram	31	480...	4

Order_by

```sql
SELECT *
FROM Employee
ORDER BY Salary DESC;
```

Em...	Emp_...	Age	Salary	Dept_ID
107	Ravi	32	600...	5
105	Kiran	30	550...	4
103	Arjun	29	500...	2
109	Vikram	31	480...	4
102	Priya	27	450...	2
108	Meena	28	420...	3
106	Ananya	26	350...	1
101	Rahul	25	300...	1
104	Sneha	24	280...	3
110	Divya	23	250...	5

Group_by

```sql
SELECT
Dept_ID,
AVG(Salary) AS Average_Salary
FROM Employee
GROUP BY Dept_ID;
```

Dept_ID	Average_Salary
1	32500
2	47500
3	35000
4	51500
5	42500

Aggregate functior

```sql
SELECT
MAX(Salary) AS Highest_Salary,
MIN(Salary) AS Lowest_Salary,
COUNT(*) AS Total_Employees,
SUM(Salary) AS Total_Salary,
AVG(Salary) AS Average_Salary
FROM Employee;
```

Hig...	Lowes...	Total...	Total...	Average_S...
600...	250...	10	418...	41800

Inner join

```sql
SELECT
    Employee.Emp_ID,
    Employee.Emp_Name,
    Department.Dept_Name,
    Department.Location
FROM Employee
INNER JOIN Department
ON Employee.Dept_ID = Department.Dept_ID;
```

Emp_ID	Emp_Na...	Dept_Na...	Location
101	Rahul	HR	Hyderabad
102	Priya	IT	Bangalore
103	Arjun	IT	Bangalore
104	Sneha	Finance	Chennai
105	Kiran	Marketing	Mumbai
106	Ananya	HR	Hyderabad
107	Ravi	Sales	Delhi
108	Meena	Finance	Chennai
109	Vikram	Marketing	Mumbai
110	Divya	Sales	Delhi

```sql
SELECT
    Employee.Emp_Name,
    Department.Dept_Name
FROM Employee
LEFT JOIN Department
ON Employee.Dept_ID = Department.Dept_ID;
```

Emp_Name	Dept_Name
Rahul	HR
Priya	IT
Arjun	IT
Sneha	Finance
Kiran	Marketing
Ananya	HR
Ravi	Sales
Meena	Finance
Vikram	Marketing
Divya	Sales

```sql
SELECT
    Employee.Emp_Name,
    Department.Dept_Name
FROM Employee
RIGHT JOIN Department
ON Employee.Dept_ID = Department.Dept_ID;
```

Emp_Name	Dept_Name
Rahul	HR
Priya	IT
Arjun	IT
Sneha	Finance
Kiran	Marketing
Ananya	HR
Ravi	Sales
Meena	Finance
Vikram	Marketing
Divya	Sales

Sub_querry

```sql
SELECT *
FROM Employee
WHERE Dept_ID IN
(
    SELECT Dept_ID
    FROM Department
    WHERE Location IN ('Hyderabad',
                       'Bangalore')
);
```

Em...	Emp_...	Age	Salary	Dept_ID
101	Rahul	25	300...	1
102	Priya	27	450...	2
103	Arjun	29	500...	2
106	Ananya	26	350...	1

Groupby, where, having

```sql
SELECT
    Dept_ID,
    COUNT(*) AS Employee_Count
FROM Employee
WHERE Salary > 30000
GROUP BY Dept_ID
HAVING COUNT(*) > 1;
```

Dept_ID	Employee_Count
2	2
4	2

View

```sql
CREATE VIEW Employee_Details AS
SELECT
    Employee.Emp_ID,
    Employee.Emp_Name,
    Employee.Salary,
    Department.Dept_Name,
    Department.Location
FROM Employee
INNER JOIN Department
ON Employee.Dept_ID = Department.Dept_ID;
SELECT * FROM Employee_Details;
```

Em...	Emp_...	Salary	Dept_...	Location
101	Rahul	300...	HR	Hyderabad
102	Priya	450...	IT	Bangalore
103	Arjun	500...	IT	Bangalore
104	Sneha	280...	Finance	Chennai
105	Kiran	550...	Market...	Mumbai
106	Ananya	350...	HR	Hyderabad
107	Ravi	600...	Sales	Delhi
108	Meena	420...	Finance	Chennai
109	Vikram	480...	Market...	Mumbai
110	Divya	250...	Sales	Delhi

Department_analys

```sql
SELECT
    Department.Dept_Name,
    AVG(Employee.Salary) AS Average_Salary
FROM Employee
INNER JOIN Department
ON Employee.Dept_ID = Department.Dept_ID
GROUP BY Department.Dept_Name
ORDER BY Average_Salary DESC
LIMIT 1;
```

Dept_Name	Average_Salary
Marketing	51500